Digital Resource Companion · V1.0

Artificial Intelligence and Healthcare Economics

Evidence, Value, Governance, and Sustainability

This guide provides the simplest entry into the comprehensive companion. Use the web layer or guided pathways before opening the raw canonical folders.

Open the Web Companion

Choose a decision pathway

Templates and workbooks

Choose your goal

Learn the approach

Read the aligned book chapter, open a worked example, and use the Common Case-Analysis Method.

Make an institutional decision

Select one of the seven pathways below and begin with its essential resource set.

Use a template or workbook

Use the browser forms or consolidated DOCX/ODT/PDF and XLSX/ODS bundles.

Run an analysis

Use the Python Analytics guide only when the decision requires quantitative analysis.

Teach or facilitate

Use synthetic cases, reference guides, and the structured case-analysis method.

Access advanced source

Open raw Chapter folders, process source, schemas, Python, JSON, and LaTeX.

Guided pathways

Define a new proposal

Decision question: What service problem is being addressed, compared with what, and is the institution ready to proceed?

Essential resources: AIHE-A01, AIHE-A02, AIHE-A03, AIHE-A04

Expected output: A problem-first proposal, comparator set, readiness position, evidence gaps, and a proceed, remediate, pilot, or stop decision.

Prepare an economic and affordability case

Decision question: Do expected outcomes justify full lifecycle cost and budget exposure, and which uncertainties could change the decision?

Essential resources: AIHE-B01, AIHE-B02, AIHE-B03, AIHE-B04, AIHE-B05

Expected output: A transparent cost, affordability, value, uncertainty, and evidence-priority record.

Validate a system locally

Decision question: Does the exact system and version perform acceptably in the intended local population, workflow, and decision window?

Essential resources: AIHE-A04, AIHE-EX05, AIHE-B05, AIHE-C01

Expected output: A local-validation report with limitations, subgroup findings, clinical utility, monitoring conditions, and decision recommendation.

Evaluate and contract with a vendor

Decision question: Which solution and contract best meet the defined need while preserving evidence access, monitoring, continuity, and exit?

Essential resources: AIHE-A01, AIHE-B02, AIHE-C03, AIHE-D01, AIHE-D02

Expected output: A due-diligence record, vendor comparison, contractual requirements, approval gates, and negotiation position.

Implement and monitor safely

Decision question: How should the intervention be introduced, monitored, corrected, and scaled without losing control of safety, workload, equity, or value?

Essential resources: AIHE-C01, AIHE-C02, AIHE-D02, AIHE-D03, AIHE-D04

Expected output: A staged implementation, monitoring, incident, corrective-action, and scale decision record.

Review an update, incident, or renewal

Decision question: Does new evidence, an incident, a material change, or renewal justify continuation, redesign, restriction, replacement, or retirement?

Essential resources: AIHE-C02, AIHE-D03, AIHE-D04, AIHE-D05

Expected output: A reconstructed event or change record and a documented continue, restrict, replace, or retire decision.

Assess circularity and asset stewardship

Decision question: How can health-system value be preserved while reducing avoidable demand, extending safe useful life, and managing environmental and material consequences?

Essential resources: AIHE-E01, AIHE-E02, AIHE-E03, AIHE-E04, AIHE-E05

Expected output: A circularity, procurement, asset-option, sustainable-value, and disposition decision record.

Before institutional use

> Create a governed local copy. Replace synthetic assumptions with current, authorized local evidence. Record the exact system and version, data period, evidence cut-off, accountable owner, uncertainty, safeguards, monitoring thresholds, next review date, and exit route. Do not place identifiable patient information, credentials, private keys, confidential contracts, or uncontrolled data in public or ungoverned copies.

Reader levels

- **Introductory:** orientation, worked examples, and reference guides.
- **Applied:** institutional records requiring local evidence and multidisciplinary review.
- **Advanced:** modelling, validation, uncertainty, audit, and specialist methods.
- **Technical source:** Python, Jupyter, schemas, LaTeX, Mermaid, and metadata.

For the complete resource inventory, open the resource browser and select the advanced view only when needed.